

AI SURGEON

See one. Do one. Teach one.

A phone-first surgical training product where you cannot cut something you cannot name.

Concept and screen design · August 2026

Companion to the business plan · Prepared for development partners

An AI attending performs the operation, names every structure as it appears, then hands over the instruments and supervises. It asks questions mid-case and points ride on the answer. Before a learner is allowed to scrub in, they have to study the anatomy and the pathophysiology and get it right.

PHASE 1

Phone. Browser-based.
No install. Touch gestures.

PHASE 2

VR. Same codebase.
WebXR, already wired.

PHASE 3

ProVR Bluetooth instruments.
Optional, like a stylus.

THE PRODUCT, ON A PHONE

Identify before you cut. Call for your instruments by name. Answer the attending mid-case. Then teach it to someone junior to you — the phase worth the most points, and the one nothing else on the market tests.



The incumbents already bought the expensive ground.

Osso VR owns headset-based institutional procedural training. Level Ex owns pharma-funded gamified specialty content. Touch Surgery, owned by Medtronic, already puts procedural rehearsal on a phone and gives it away.

None of them owns the layer beneath the procedure, which is anatomy a learner actually has to know rather than click through. That is the opening, and it is the only ground worth entering on.

This document is not a request for \$500M to build a AAA title. The phone version with four to six modules is a small-team build — one developer, one anatomy artist, one surgical content reviewer, nine to twelve months.

THE ONE DESIGN DECISION

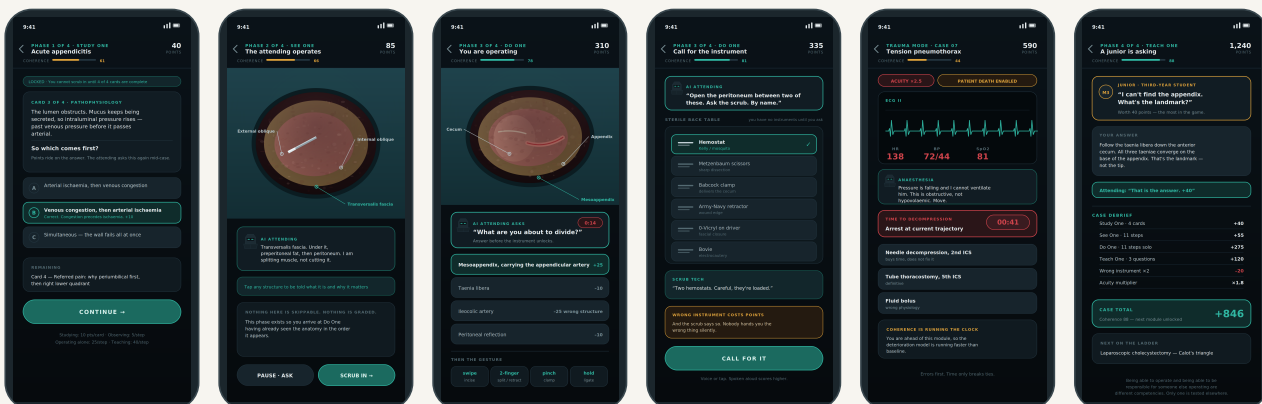
The loop does not open on a scalpel. It opens on the appendix — what it is, what it does, why this one has to come out.

Then the pathophysiology. The lumen obstructs, mucus keeps being secreted, pressure passes venous before arterial, so congestion precedes ischaemia. Then bacterial translocation, gangrene, perforation.

Four cards. Four questions. Ten points each. You cannot scrub in until you have done them.

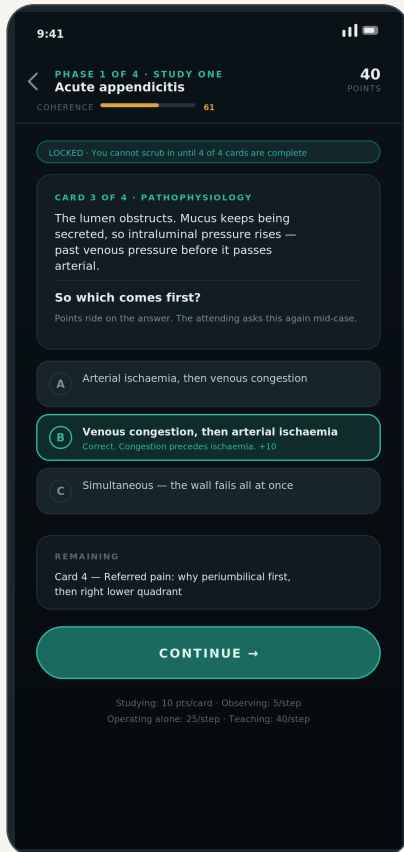
The game will not let you cut something you have not studied.

THE LOOP, IN SIX SCREENS



Screen design is final-intent. Anatomy shown is placeholder — see the note on sourced anatomy, page 7.

First you study. Then you watch an attending do it properly.



STUDY ONE · 10 POINTS A CARD

Pre-op anatomy and physiology, gated. The card above is the one every examiner asks about: pressure rises past venous before arterial, so venous congestion precedes arterial ischaemia. Get it wrong and you review it. Get it right and the attending will still ask you again mid-case.

SEE ONE · 5 POINTS A STEP

Not a cutscene. The attending performs the procedure and names every structure as it appears. The learner can stop at any moment, tap anything on screen, and be told what it is and why it matters clinically.

Nothing here is skippable and nothing here is graded. The phase exists so the learner arrives at the next one having already seen the anatomy in the order it appears.

ANATOMICAL CONTENT STANDARD

Every structure carries a name, a line on what it is, and a line on why it matters clinically. Every step sequence has to match an accepted operative technique and be reviewable against a standard text. Where technique is genuinely contested — stump inversion, for instance — the module says so rather than picking a side silently. This is the part that cannot be outsourced. It is also the moat: a competitor can copy the gesture system in a quarter.

THE LOOP, AND THE GATE THAT RUNS IT

STUDY ONE

Anatomy and pathophysiology.
Gated. 10 a card.

→ SEE ONE

The attending operates and
names everything. 5 a step.

→ DO ONE

You identify, then you act.
25 a step, alone.

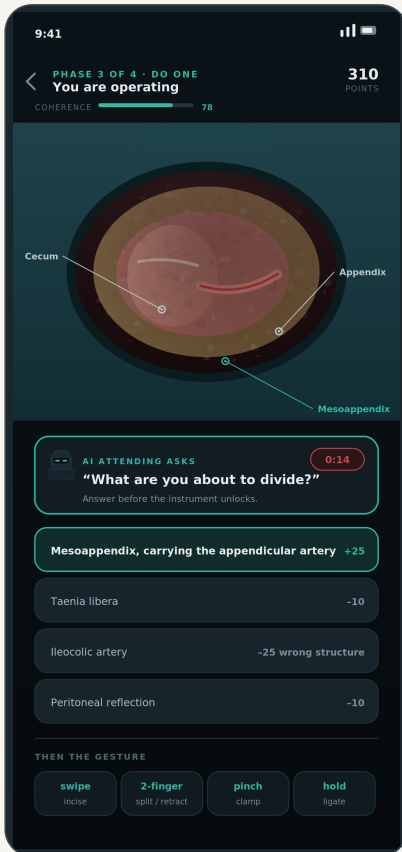
→ TEACH ONE

A junior asks. You answer.
40 a step.

COHERENCE GATE Above 72, the next module opens at the difficulty you left off at.

34 to 72 repeats Do One. Below 34 sends you back to the specific structures you missed, then back to See One.

You do not get to cut something you cannot name.



IDENTIFY, THEN ACT

The attending calls the step. The learner must identify the correct structure before the maneuver unlocks. Then the gesture: swipe to incise, two-finger spread to split bluntly or retract, pinch to clamp, press-and-hold to ligate.

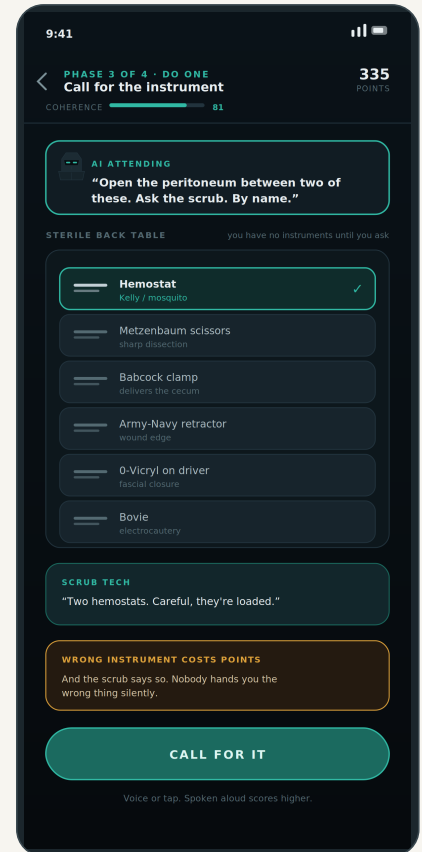
Wrong structure, wrong maneuver, and wrong-place-right-maneuver are three different errors and are scored differently, because they are three different kinds of not knowing.

THE SCRUB TECH

You do not have instruments. You ask for them, by name, and the scrub hands them over.

Knowing the peritoneum is opened between two hemostats, that the external oblique aponeurosis is taken with Metzenbaums, that the cecum is delivered with a Babcock and not a Kelly — that is real operative fluency, and no other product makes you produce it from memory.

It also teaches the social choreography of an OR, which students are terrified of and no simulator addresses. Wrong instrument costs points and the scrub says so. Nobody hands you the wrong



THE ECONOMY IS DELIBERATELY LOPSIDED

10

study, per card

5

observing, per step

25

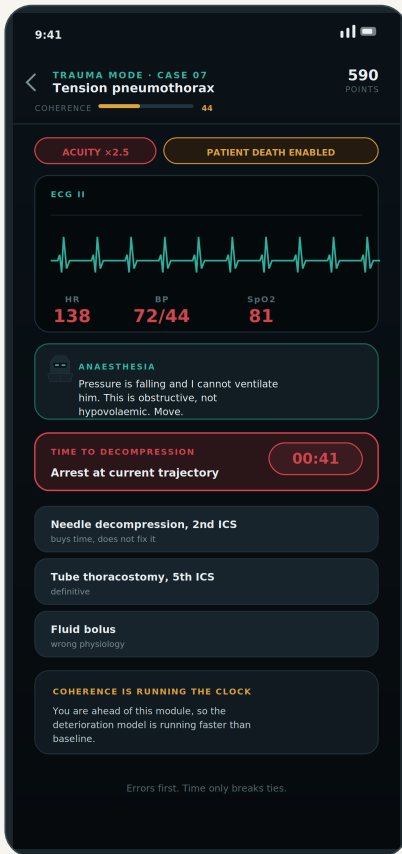
operating alone

40

teaching a step back

Acuity multiplies reward and penalty together. A routine nineteen-year-old is $\times 1.0$. A perforated appendix with an abscess in a diabetic is $\times 1.8$. A septic seventy-year-old on anticoagulation is $\times 2.5$. Taking a hard case is a real gamble, not a free points grab.

Trauma is a mode, not a module.



A randomized case, a clock, incomplete information, and an anatomy you have to identify under time pressure with the field partly obscured.

LIVE PHYSIOLOGY, NOT A TIMER

The monitor is driven by a deterioration model, not a countdown. In the built tension pneumothorax module, doing nothing arrests the patient in 56 to 78 seconds. A needle buys 67 to 110. The tube has to be in within 63 to 103. The spread depends on how well the learner is doing.

COHERENCE MONITORING

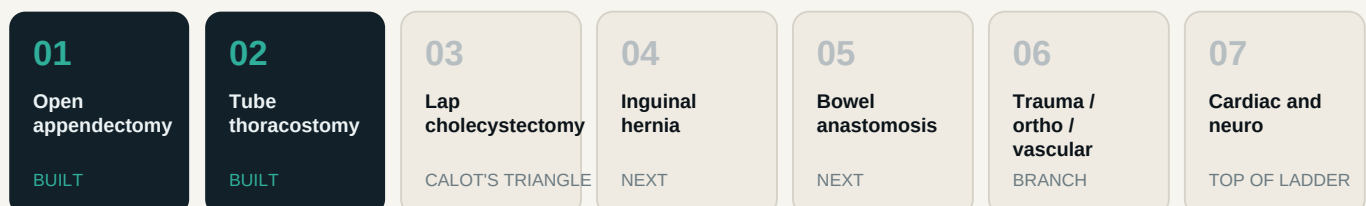
The system tracks accuracy and hesitation continuously and moves difficulty in both directions during play, not between sessions. High coherence and the labels stop appearing, gesture tolerance tightens, the attending stops narrating, the teaching questions get harder — and the clock runs faster. Low coherence and the scaffolding quietly comes back.

A learner who is struggling should not be failed out. A learner who is ahead of the module should feel it get harder within thirty seconds, not at the next unlock.

AND YOU CAN LOSE THE PATIENT

A death voids the case score and takes a fixed penalty off rank. It must be rare, always traceable to specific decisions the learner made, and followed by a debrief that walks through exactly where it went wrong. Handled well it is the most powerful teaching moment in the product. Handled badly it is cruel and people quit. It is not possible at all in the entry modules.

THE RESIDENCY LADDER · ORDERED BY ANATOMICAL DEMAND, NOT BY DRAMA



Teach one is the phase that makes the whole thing worth building.

A junior asks a question and the learner has to answer it. Not “what is the next step” but “why is the incision oblique,” “trace this artery back to the aorta,” “your junior cannot find the appendix, what is the landmark.”

Being able to perform a procedure and being able to be responsible for someone else performing it are different competencies. Almost nothing on the market tests the second one. It is worth the most points in the game — forty a step, against twenty-five for operating alone — because it is the harder thing.

COMPETITION · THREE FORMATS ON ONE CASE SEED

Same team, same case

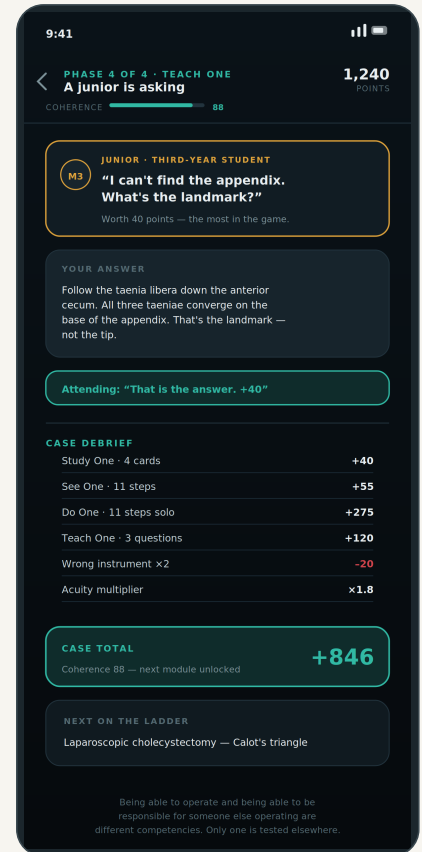
Two to four players in one OR with real roles — primary surgeon, first assist, scrub, and someone running the anaesthetic. The surgeon calls for an instrument and a human being has to hand it over. This is the format that will actually get played, because it recreates the thing that makes an OR interesting: a team performing under pressure.

Head to head

Same patient, same seed, separate runs. Fewest errors wins; time breaks the tie. Errors first and time second, always, in that order. If speed wins outright you have built a game that teaches people to rush, which is the exact opposite of the intent.

Ladder

Seasonal ranking across a school or a program, acuity-weighted so the leaderboard rewards people taking hard cases well rather than farming easy ones fast.



THE AI ATTENDING IS A CHARACTER

It should read as imposing, precise, and worth impressing — armored, mechanical, deliberate in how it moves and speaks. One practical caution: build it as an original design. A character that reads as a specific existing film robot is a trademark and copyright exposure that would follow the product into every funding conversation and every app store review, and it is entirely avoidable at the concept-art stage. The silhouette can carry the same weight without borrowing anyone's.

Build notes and honest costs.

SOURCED ANATOMY, NOT MODELLED ANATOMY

This is the single most important production decision in the project, and the first prototype got it wrong. Anatomy built from geometric primitives — a sphere for a cecum, a torus for a taenia — fails on sight with the exact audience the product needs, and no engine choice fixes it.

Anatomy must be sourced, not authored. Three viable routes:

- Z-Anatomy / BodyParts3D — an open 3D atlas of over 7,000 labelled structures derived from real scan data, exportable to glTF. Licensed CC BY-SA, which is copyleft; whether share-alike encumbers a commercial product built around it is a question for counsel before committing.
- NLM Visible Human Project — public-domain cryosection, CT and MRI data. No license required since 2019. Photographs of an actual human, which is why nothing rendered competes with it for the study cards.
- Real operative video for See One — surgeons already learn this way, so it is the format the audience trusts by default. SURGhub, the UN's surgery learning hub, publishes a free acute appendicitis course with appendectomy footage.

PLATFORM

Phase 1 is one web build running in the browser on iOS and Android with no install. It has to be genuinely good on its own; it is not a demo for a VR product. Phase 2 is the same codebase with WebXR — short trigger pull identifies, long pull performs. Phase 3 is Bluetooth instrument replicas: optional, like a stylus, and last because it is hardest

WHAT EXISTS TODAY

Two working browser modules — an open appendectomy via McBurney gridiron incision, and a tube thoracostomy for tension pneumothorax. The second adds a live vitals monitor driven by an obstructive-shock model, an anaesthesia character who calls out physiology, and patient death as a real fail state. Both are anatomy-fact-checked and both have been driven headlessly by a test harness that checks for softlocks, exact scoring and arrest timing.

Modules are data — layers, structures, brief, steps, tray — sitting on a shared engine, but the engine is copy-pasted per module rather than extracted. Extract it before module four.

WHAT THIS HAS TO PROVE FIRST

- 1 Does identify-before-you-cut teach anatomy better than a labelled diagram? Testable with a pre/post quiz and forty students, for almost nothing.
- 2 Does Teach One retain people? It is the differentiated mechanic and the untested one.
- 3 Will a surgical educator put their name on the content? One named academic surgeon reviewing the general surgery track is worth more than the entire market-size section of the old plan.

WHERE THE REWARDS COME FROM

The endpoint — real merchandise, real scholarships — stays. Who funds it changes. The company funding scholarships out of a raise makes the rewards a cost centre and the whole thing a liability. Turn it around: schools, nursing programs, hospital systems, and state or federal STEM grant programs fund the incentives; the platform administers them and takes a licensing fee. A nursing school that will pay for a leaderboard prize pool for its own cohort is a far easier first customer than a sovereign wealth fund, and it is reachable next semester rather than in five years. Start concrete: a hospital system sponsors a set of loupes for the top resident in a quarter. A nursing program funds a \$500 award. None of that requires a raise, and all of it builds the evidence the scholarship argument needs at scale.

The game will not let you cut something you have not studied.

Everything else in the design follows from that one sentence.

CONTACT

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A NOTE ON THIS DOCUMENT

This brochure follows the August 2026 scoped concept, which supersedes the earlier master business plan. Screen designs are final-intent; the anatomy shown in them is placeholder, and the production art must come from the sourced datasets named on the previous page rather than be modelled from scratch.

Market figures from the original business plan are deliberately absent here. Several of them — the VR-medical-training market size, the “37% better” laparoscopic claim, and the 2025 to 2027 revenue projections — need re-verification against primary sources or removal before any version of this goes to an outside reader. Nothing in this document depends on them.